

PAPER_892: SCm Kozima Phonon Resonance Coupling

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Abstract

Kozima phonon resonance coupling at 1.25 THz within the SCm vacuum field. $F_{\text{neutron}} = n \cdot \sigma_n(\omega) \cdot v_{\text{th}} \cdot \hbar \omega$ with Gaussian cross-section σ_n peaking at the SCm resonance frequency. Drives LENR neutron drop reactions at lab-accessible THz frequencies.

1. Core Equations

$$\sigma_n(\omega) = \sigma_0 \cdot \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)]$$
$$F_{\text{neutron}} = n \cdot \sigma_n \cdot v_{\text{th}} \cdot \hbar \omega$$

2. Parameters

Parameter	Default	Description
omega	$2\pi \times 1.25\text{e}12$ rad/s	Angular frequency
T	300 K	Temperature
n_density	$1\text{e}28$ m ⁻³	Number density

3. UQFF Integration

This calculator integrates `et_scm_vacuum.py` from Session 207 into the CP4 pipeline as class #476.

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